Docker Commands & Practical Notes

Compiled from the uploaded pages

S.No.	Docker Command / Topic	Meaning / Use
1	Docker info	Shows Docker system information.
2	docker images	View list of all Docker images.
3	docker run <image-name>	Create and run a container from an image.
4	docker ps -a	Check list of all containers.
5	docker run -it <image-name>	Run container in interactive (-i) + terminal (-t) mode.
6	ls	Check all directories.
7	mkdir <directory-name>	Make a new directory.
8	docker stop <container-name> docker stop <container-id>	Stop a running container.
9	docker start <container-name> docker start <container-id>	Start an existing stopped container.
10	docker pull <image-name>:<image-version> Example: docker pull mysql:8.0	Download/pull an image from Docker registry.
11	docker run -d <image-name>	Run image in detached mode (background).
12	docker run --name <container-name> -d <image-name>	Give the container a custom name and run in detached mode.
13	docker rmi <image-name> docker rmi <image-id>	Remove an image.
14	docker rm <container-name> docker rm <container-id> kisi Docker image k sabhi container ek sath remove kar ho to . docker rm -f $(docker ps -a -q --filter "ancestor=ubuntu")	Remove a container.
15	docker run -d --name <container-custom-name> -e <container-current-name> Example: docker run -d --name tarunsql -e MYSQL_ROOT_PASSWORD=1234 mysql custome name docker run -d --name tarunsql mysql:latest -e MYSQL_ROOT_PASSWORD=1234 mysql docker logs tarunsql	Define an environment variable. Common MySQL example: MYSQL_ROOT_PASSWORD.

	docker rm <container Id>
	
	Ek se zyada environment variables bhi de sakte ho:
	
	Bash
	`docker run -d --name tarunsql ^`
	`-e MYSQL_ROOT_PASSWORD=1234 ^`
	`-e MYSQL_DATABASE=college ^`
	`-e MYSQL_USER=tarun ^`
	`-e MYSQL_PASSWORD=abc123 ^`
	`mysql`	

16	Port Binding docker run -p 8080:3306 <image-name> Format: hostPort:containerPort	Map a host port to a container port.
17	docker run -d --name mysql-latest -p 8080:3306 -e MYSQL_ROOT_PASSWORD=secret mysql:8.0 docker run -d --name shivamsql -p 8080:3306 -e MYSQL_ROOT_PASSWORD=12345 mysq	Example combining port binding, name, detached mode and environment variable.
18	docker logs <container-id>	Troubleshooting: view container logs.
19	docker exec -it <container-id>/bin/bash docker exec -it <container-id>/bin/sh	Open an interactive shell inside a running container.
20	docker network ls	Fetch/list all Docker networks.
21	docker network create <network-name>	Create a Docker network.
22	docker run -d ^ -p 27017:27017 ^ -e MONGO_INITDB_ROOT_USERNAME=tarun ^ -e MONGO_INITDB_ROOT_PASSWORD=tarun2805 ^ --name tarunMongo ^ --network mongo-network ^ mongo	Run a MongoDB container with port binding, root credentials, custom name and a user-defined network.
23	Docker Compose	Used to set up multiple/container-based services and their configuration.
24	docker-compose -f <filename.yml> up -d	Start services in detached mode.
25	docker-compose -f <filename.yml> down	Stop/remove the Compose containers.
26	docker network ls	Useful Docker checking,

	docker ps docker logs <container-id> Docker docs / Read docker file docker hub search <image>	troubleshooting and documentation commands.
27	docker build -t tarunapp: 1.4 docker run ^ -p 3000:3000 -e ^ MONGO_URI="mongodb://tarun:tarun2805@host.docker.internal: 27017/?authSource=admin" tarunapp:1.4	Build an image from a Dockerfile and run the created image.
28	Push the image on Docker Hub docker build -t tarun2805/nodeapp . docker login docker push tarun2805/nodeapp docker logout docker pull tarun2805/nodeapp:latest	Build, log in, push an image to Docker Hub, log out, and pull the published image.
29	Docker volume docker run -it -v "C:\Users\Tarun2\Desktop\datafolder:/test/data" ubuntu docker exec -it 62c0f892fccd /bin/bash	Persistent storage for Docker containers.
30	docker run -v volume_Name:<container_dir_path>	Named volume: mount a named Docker volume into a container.
31	docker run -v <mount_path>	Anonymous volume: Docker creates/manages the volume automatically.
32	docker run -v <host_dir_path>:<mount_dir_path>	Bind mount: map a host directory to a container directory.
33	Docker network drivers: 1. Bridge — default network driver 2. Host 3. Null	Common Docker network driver types.

🔥 **Sabse important 6 yaad rakho:**

```
0   → Success
1   → General Error
126 → Command execute nahi hui
127 → Command nahi mili
137 → SIGKILL / often OOM
139 → Segmentation Fault
143 → SIGTERM
```

Aur **exact error jaanne ke liye** `docker logs <container>` **sabse useful command hai.**

```
Exited (0)   → normally successfully finished
Exited (1)   → error/failure
Exited (137) → process was killed (often SIGKILL/OOM)
Exited (255) → abnormal/error exit
```

dikhte hain.

1. Pehle `STATUS` ko samjho

```
docker ps -a
```

Example:

```
STATUS
Up 10 seconds
Exited (0) 2 minutes ago
Exited (1) 5 minutes ago
Created
Restarting (1) 10 seconds ago
Paused
Dead
```

Ye sab **container states/statuses** hain.

2. Docker Container ki main states

State	Meaning	Example
Created	Container create ho gaya, but start nahi hua	`Created`
Running / Up	Container ka main process currently running hai	`Up 2 minutes`
Exited	Main process terminate ho gaya	`Exited (0)`
Paused	Container ke processes temporarily freeze/suspend hain	`Paused`
Restarting	Docker container ko automatically restart kar raha hai	`Restarting (1)`
Dead	Container terminate hua, lekin Docker proper cleanup nahi kar paya	`Dead`
Removing	Container remove hor process mein hai	`Removing`

```
                 Docker
                   │
         ┌─────────┴────────────┐
         │                      │
     tarunmongo             mongo-express
     MongoDB                Web UI
     :27017                 :8081
         ▲                      ▲
         │                      │
   localhost:27017        localhost:8081
     DB protocol               HTTP
         │                      │
       ❌ Chrome              ✅ Chrome
```